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Bihar Engineering University, Patna

B.Tech 3rd Semester Exam., 2025

Code: 103302 | ***Time: 03 Hours***

Semester: III | ***Subject: Analog Electronics*** | ***Full Marks: 70***

Instructions:

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct option / answer the following (Any seven questions only) **[2 x 7 = 14]**

(a) For a phase shift oscillator with equal RC sections, frequency is

(i) $1/2\pi RC$ (iii) $1/(2\pi RC\sqrt{6})$
(ii) $1/\pi RC$ (iv) $1/RC$

(b) In a Zener regulator, maximum power dissipation occurs when

(i) The Load current is maximum (iii) The Load current is minimum
(ii) Input voltage is minimum (iv) Zener is off

(c) An 8-bit ADC with a 5V reference has a resolution

(i) 19.53 mV (iii) 5 mV
(ii) 10 mV (iv) 2.5 mV

(d) In BJT, high-frequency response is limited by

(i) Resistances (iii) Capacitances
(ii) Inductance (iv) Power

(e) A silicon diode at 300K carries 1 mA at 0.7V. The current at 0.8V is approximately

(i) 2 mA (iii) 10 mA
(ii) 5 mA (iv) 20 mA

(f) In a PN junction under forward bias, the depletion region

(i) Widens (iii) Remains same
(ii) Narrows (iv) Breaks

(g) MOSFET, in saturation region

(i) $V_{DS} < V_{GS} - V_T$ (iii) $V_{GS} < V_T$
(ii) $V_{DS} > V_{GS} - V_T$ (iv) None

(h) In reverse bias, diode behaves like

(i) Short circuit (iii) Capacitor
(ii) Open circuit (iv) Inductor

(i) MOSFET acts as current source in

(i) Cutoff (iii) Saturation
(ii) Triode (iv) Breakdown

(j) In an op-amp circuit, slew rate limits

(i) Gain (iii) Rate of output change
(ii) Frequency response (iv) Offset

Q.2 **[7 x 2 = 14]**

(a) Explain the current mirror circuit and its applications.

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(b) Explain the instrumentation amplifier and its advantages.

Q.3 **[7 x 2 = 14]**

(a) Explain the working of a bridge rectifier with waveform diagrams.

(b) Design a square-wave generator using an op-amp and explain its operation.

Q.4 **[7 x 2 = 14]**

(a) Explain the small-signal model of MOSFET.

(b) Derive the expression for the voltage gain of a non-inverting amplifier.

Q.5 **[7 x 2 = 14]**

(a) Calculate the input and output impedance of a CE BJT amplifier.

(b) Evaluate the limitations of differential amplifiers in practical applications.

Q.6 **[7 x 2 = 14]**

(a) Describe the basic working principle of a differential amplifier.

(b) Explain the structure and operation of MOSFET.

Q.7 **[7 x 2 = 14]**

(a) Design a PI controller using an op-amp and explain its advantage over P controller.

(b) Explain the operation of a clipping circuit and design a positive clipper.

Q.8 **[7 x 2 = 14]**

(a) Derive expressions for upper and lower threshold voltages in a Schmitt trigger by taking a suitable circuit diagram.

(b) Examine the impact of input bias current on op-amp circuits.

Q.9 Write short notes on any two of the following: **[7 x 2 = 14]**

(a) Hybrid-π model of BJT

(b) Waveform shaping circuits using op-amp

(c) Need for filters in rectifier circuits

(d) Miller's theorem

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